

PULSAR: Pooled Unified Late-Interaction Search and Retrieval for Enterprise Visual Document RAG

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Abstract

Institutional investors search visually dense pitch decks, board packs, and diligence materials that change hourly near deal closing. OCR followed by figure verbalisation is costly to refresh at this scale and can lose chart detail. We present PULSAR, a production vision-first retrieval system deployed at Mubadala Investment Company. PULSAR indexes page images with a frozen ColPali-style backbone and uses a pooled two-stage late-interaction index: compact page summaries support initial retrieval, followed by exact MaxSim rescoring over a finer pooled representation. On ViDoRe V3, this design reduces median vector-search latency by 15.1 times against an unpooled configuration with less than 0.01 absolute NDCG@10 and Recall@10 loss; production median vector-search latency is 156 ms. Under concurrent load, the pooled index sustains approximately 88 times higher QPS than an unpooled index. The event-driven ingestion path is estimated to be approximately 20 times cheaper per page than the OCR+verbalisation baseline it replaced. Since March 2026, PULSAR has served 78 thousand documents and approximately 2.4 million pages across more than 3,000 deals. At the production top K, it more than doubles answer-fact recall over the OCR+verbalisation baseline.

1 Introduction

Enterprise RAG (Lewis et al., 2020) is harder than text-only benchmarks (Thakur et al., 2021): corporate decks and board packs contain tables, legends, chart annotations, and spatial cues that text-first pipelines lose even with image verbalisation. Verbalisation is information-reducing. The standard pipeline applies OCR or layout-aware extraction (Xu et al., 2020; Huang et al., 2022), chunks text, optionally adds vision-language model (VLM) captions (Liu et al., 2023; Hu et al., 2024), and retrieves with hybrid search. Each page is run

through a document OCR+layout model and a per-figure VLM, then the verbalised text is embedded: three passes where vision-first uses one. At firm scale (~ 2.4 M chart-heavy pages), this is an order of magnitude more work per page than a single embedding pass and is repaid every time a deal document changes near closing.

Vision-first inverts this. The page image becomes the unit of indexing and grounding. Ingestion collapses to a single forward pass of the retriever’s own embedding model per page, dropping the separate OCR+layout and per-figure captioning passes and making cost scale with page count alone. On enterprise corpora, Loison et al. (2026) report visual retrievers outperforming textual ones.

We build on VLM-based multi-vector page retrievers with late-interaction scoring (Khatab and Zaharia, 2020; Faysse et al., 2025), rather than OCR-free document transformers (Kim et al., 2022; Hu et al., 2024). Late interaction keeps many embedding vectors per page (one per image patch) rather than collapsing the page to a single vector, so chart values and layout cues stay addressable. At firm scale this puts billions of vectors in the index, and per-query scoring cost grows linearly in patches per page.

PULSAR (Pooled Unified Late-interaction Search And Retrieval), our production retrieval system, is built around this cost. It composes a frozen ColPali-style backbone with a pooled two-stage retrieval pipeline and an event-driven ingestion path. Our contribution is not a new retrieval primitive but the engineering and empirical work of scaling this vision-first paradigm to production at firm scale.

Contributions. (i) A decoupling of embedding-time and answer-time resolution: 150 DPI suffices for retrieval, while 500 DPI at answer time drives page-legibility errors on firm decks to zero. (ii) A two-stage pooled late-interaction index that combines mean row/column pooling (Qdrant, 2026) for the first-stage lookup with hierarchical-pooled ex-

act rescoring (Clavié et al., 2024) and binary quantisation, selected by a pooling-and-quantisation sweep. It cuts single-stream median vector search latency $15.1\times$ at under 0.01 NDCG@10 and Recall@10 loss, and holds a 156 ms median under production concurrency. (iii) An ingestion recipe (rolling rasterisation, strict-priority work queues, page-hash resume) that scales with page count rather than visual complexity. It is $\approx 20\times$ cheaper per page than the OCR+verbalisation baseline it replaces (call-count estimate, §5.4). (iv) A production deployment at Mubadala Investment Company (78k documents across >3,000 deals, $\sim 2.4\text{M}$ pages, live since March 2026). An in-house end-to-end answer evaluation shows PULSAR exceeding an OCR+verbalisation baseline on context-fact recall, answer-fact recall, and completeness at every top- k .

2 Related Work

Vision-first document retrieval. Late interaction began with ColBERT (Khattab and Zaharia, 2020; Santhanam et al., 2022b), which gives each document one vector per token instead of a single dense vector and scores a query–document pair by MaxSim: for each query token, take its best match against any document token and sum. The index grows, but matching becomes finer-grained. ColPali (Faysse et al., 2025) carries the idea to page images. A vision-language backbone (Wang et al., 2024) splits the page into patches, each patch becomes a token, and MaxSim then runs between query text tokens and image patches.

Generate-and-encode hybrids. The alternative to indexing the image is to verbalise it first. Systems such as SERVAL (Nguyen et al., 2025) verbalise page images before text-encoding and themselves flag VLM-generated text as an unmeasured hallucination source. Both paradigms run a VLM, but verbalisation bakes its output into the index at ingestion; vision-first keeps the page image as the stored unit and defers generation to answer time, grounded on the retrieved page. Verbalisation often captures the concrete values on a chart, but not how the items relate visually: the groupings and orderings a reader uses to interpret it. Some of what analysts query for is lost in that gap.

Generalisation to unseen corpora. The verbalisation route has one edge. Most et al. (2025) report that OCR-based RAG generalises better than vision-first retrieval *fine-tuned on a fixed corpus*.

That finding applies to fine-tuning, which we avoid: PULSAR uses a pretrained ColQwen3-4B with no firm-specific fine-tuning, and still beats the baseline on the firm’s own decks (§5.7).

Vector indexing. Dense retrieval leans on Hierarchical Navigable Small World (HNSW) (Malkov and Yashunin, 2020) with quantisation (Johnson et al., 2019) as its approximate nearest neighbour (ANN) backend. Late interaction strains that backend, since each page now carries many vectors rather than one. Prior multivector engines answer with a prefetch-then-rerank split: PLAID (Santhanam et al., 2022a) prefetches on centroids before a token-level rerank, and EMVB (Nardini et al., 2024) compresses the prefetch to bit-vectors. PULSAR keeps this split but combines two existing pooling methods across its stages: mean row/column pooling (Qdrant, 2026; Yeroyan, 2026) for the prefetch and hierarchical pooling (Clavié et al., 2024) for the exact rerank (§3.4). Both are post-hoc and need no retraining. ColMate (Masry et al., 2025) instead modifies training: its TopKSim objective averages each query token’s top- K patch similarities, reverting to plain MaxSim at inference. Recent hybrid-vector work such as HEAVEN (Kim et al., 2026) uses a single-vector first stage over visually summarised pages followed by multi-vector reranking. PULSAR instead keeps multivectors at both stages and serves them from a continuously updated production index.

3 PULSAR System Design

Late interaction buys finer-grained matching at a cost: each page becomes many patch vectors rather than one, inflating both index size and per-query scoring. PULSAR keeps this cost in check at each stage. A single backbone pass (§3.1) embeds each page at a render DPI decoupled from the higher-resolution image kept for answering (§3.3.3), so cost scales with page count, not visual complexity. Ingestion (§3.3) re-embeds only changed pages; retrieval (§3.4) scores pooled summaries rather than full patch grids, keeping exact MaxSim affordable.

3.1 Embedding backbone

PULSAR’s embedding backbone is Tomoro ColQwen3-4B (Huang and Tan, 2025), a ColPali-style multivector retriever on Qwen3-VL. At deployment, it was the strongest Apache-2.0-licensed model within our half-A10 serving budget (Table 4); the 8B sibling scored higher on ViDoRe V3

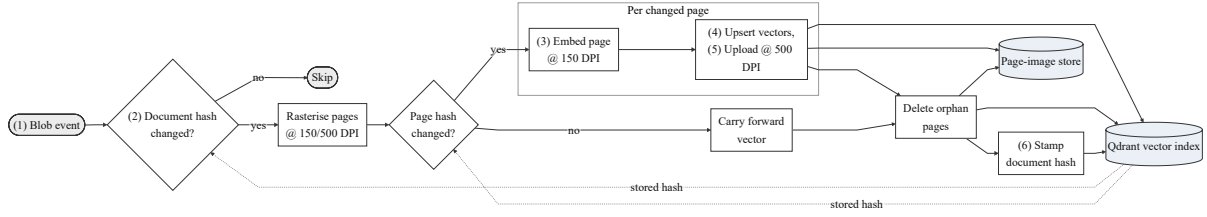


Figure 1: PULSAR’s ingestion path (§3.3): document- and page-level hashes skip unchanged work, changed pages are re-embedded with ColQwen3-4B and upserted, and the document hash is stamped once every page is durable.

(Loison et al., 2026) but exceeded that budget.

3.2 Architecture

The ingestion and retrieval paths of PULSAR¹ share two stores: a Qdrant vector index and a page-image store. Figs. 1 and 2 show the corresponding components in execution order.

Ingestion path. (1) A blob-storage event enters one of three priority queues. (2) The dispatcher takes the next document; a document-level hash skips it if unchanged, otherwise the indexer rasterises each page at 150 DPI for retrieval and 500 DPI for answer-time grounding, and a page-level hash of the render marks which pages changed. (3) ColQwen3-4B embeds the 150 DPI render of each changed page. (4) The indexer derives mean-pooled row and column vectors for HNSW prefetch and hierarchical-pooled vectors for exact reranking, then upserts them into Qdrant. (5) The 500 DPI render is written to the page-image store. (6) The document hash is stamped only after every changed page has been indexed and its page image persisted; queries filter out pages whose document has not yet been stamped.

Retrieval path. (1) ColQwen3-4B embeds the text query once and leaves its token multivectors unpooled. (2) Two HNSW searches compare the query against the mean-pooled row and column page representations. (3) Their candidates jointly form a top- k' shortlist. (4) Exact MaxSim over the hierarchical-pooled page vectors reranks that shortlist. (5) The top- $k=5$ pages are returned. (6) Their stored 500 DPI renders are fetched from the page-image store. (7) The downstream VLM reads those page images and generates the grounded answer.

¹PULSAR builds on Microsoft’s open-source Multi-Modal RAG with ColPali accelerator (Microsoft, 2026) (vLLM serving, Helm/AKS deployment); the rest of the system and the evaluation (§5) are this paper’s contributions.

3.3 Ingestion

3.3.1 Priority-aware scheduling

Three ingestion streams compete for the same GPU pool, fed to the indexer as priority lanes: high for real-time analyst uploads on a live deal, medium for hourly deal-document syncs, and low for slower syncs of non-deal material. Analyst workflows must not block behind either background stream. Scheduling splits across two layers: a custom dispatcher chooses which *document* to ingest next, and vLLM (Kwon et al., 2023) chooses which *embedding request* the GPU runs next. Both preempt lower-priority work and resume evicted jobs from the last durable page, so a burst of analyst uploads never queues behind background syncs (App. B).

3.3.2 Skip-aware re-indexing

The indexer skips redundant work at two levels (Fig. 1). A SHA-256 of raw blob bytes gives a full-document skip. A SHA-256 of each JPEG-encoded page image gives page-level skip, so re-indexing a modified document re-embeds only changed pages. Orphan pages are deleted when page count shrinks. The document hash is stamped only after every page is durable in the vector index, so the retrieval path can filter pages whose document is not yet stamped.

3.3.3 Decoupled embedding and storage DPI

We render each page *twice*: a 150 DPI render feeds ColQwen3-4B for embeddings, and a separate 500 DPI render is stored and served to the VLM at answer time. Both values come from ablations (§5.1, §5.6). Rendering runs in a bounded producer-consumer window, so the GPU stays fed while resident memory is bounded by the window, not document length.

3.4 Retrieval

ColQwen3-4B emits over 2,000 patch vectors per page, each of dimension $D=320$. At firm scale ($\sim 2.4M$ pages) this is billions of vectors live in

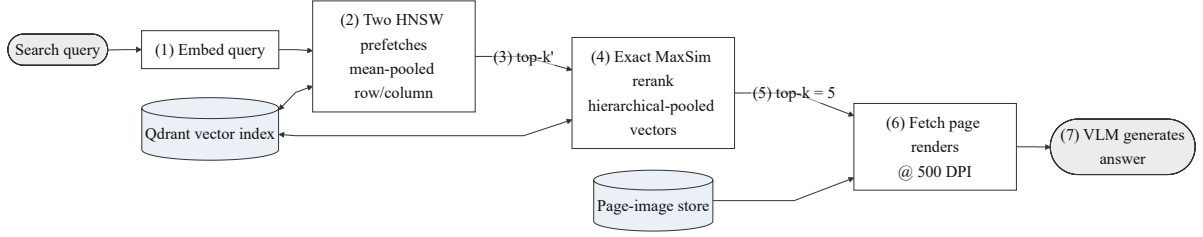


Figure 2: PULSAR’s retrieval path (§3.4): the query is embedded with ColQwen3-4B, then two quantised HNSW prefetches over mean-pooled row/column representations produce a shortlist that exact MaxSim reranks over the hierarchical-pooled vectors; the stored 500 DPI render of each returned page grounds the VLM.

HNSW (Malkov and Yashunin, 2020), and exact MaxSim costs $|q| \times |d|$ dot products per scored page. We therefore cut that count by pooling (Qdrant, 2026), and keep exact MaxSim only on a short rerank list (Santhanam et al., 2022a).

3.4.1 Pooling options

Mean row/column pooling (Qdrant, 2026) treats the per-page patch sequence as a 2D grid and averages along each axis, yielding a handful of row- and column-mean vectors of dimension D . *Hierarchical pooling* (Clavié et al., 2024) clusters similar patch vectors and replaces each cluster by its mean; the *pool factor* sets the average cluster size, so pool factor 3 retains $\sim 1/3$ of the vectors, at which Faysse et al. (2025) report 97.8% NDCG@5 retention. Mean gives a small, fixed-size summary per page; hierarchical keeps more detail at a higher per-page vector count.² These pooling methods are known individually, but no settled recipe combines them at scale, and the usual two-stage index reranks on the full multivectors. PULSAR pools that stage too, then sweeps the pooling-and-quantisation space for the fastest configuration that holds retrieval quality (§5.2).

3.4.2 Pooled two-stage retrieval

The resulting index runs two pooled stages.

Stage 1 (prefetch). Each page is summarised by 8–16 mean-pooled row/column vectors, binary-quantised in RAM (Qdrant, 2026). The query multivectors are left unpooled; two HNSW prefetches against the row-pooled and column-pooled page representations jointly produce a top- k' shortlist (Fig. 2).

Stage 2 (rerank). The shortlist is rescored with exact MaxSim over the hierarchical-pooled multivectors (Clavié et al., 2024), which sums, for each

²Other variants binarise full-patch vectors with Hamming MaxSim (Bergum, 2024), or retrain the scoring rule (Masry et al., 2025).

query text token, its best dot product against any document patch (Khattab and Zaharia, 2020):

$$s(q, d) = \sum_{i=1}^{|q|} \max_{j \in [1, |d|]} q_i^\top d_j. \quad (1)$$

The top- k are returned. Pooling also lets the full per-page patch vectors be discarded after ingestion, cutting the index’s RAM footprint about $14\times$ (§5.3).

4 Experimental Setup

The OCR+verbalisation baseline is the stack PULSAR replaced: per-page OCR+layout extraction, figure verbalisation by the query-time VLM (only filter-kept verbalisations are indexed; §5.4), semantic chunking, and dense embedding into HNSW (Malkov and Yashunin, 2020) for dense+keyword hybrid retrieval.

5 Results

We evaluate the retrieval index and its footprint (§5.1–5.3), ingestion cost and refresh lag (§5.4–5.5), then answer quality (§5.6, §5.7), where answer-fact recall more than doubles over the OCR+verbalisation baseline.³

5.1 Embedding-DPI ablation

Embedding DPI drives per-page GPU time and stored-vector count. This GPU time dominates ingestion cost at $\sim 2.4\text{M}$ pages, so we take the lowest DPI whose retrieval quality is statistically equivalent to 150 (the a-priori baseline) and let ingestion cost break the tie. We sweep DPI with a

³In every experiment the queried index has its HNSW graph verified fully built before measurement, so reported search latencies are graph-based ANN, not a brute-force scan over unindexed segments.

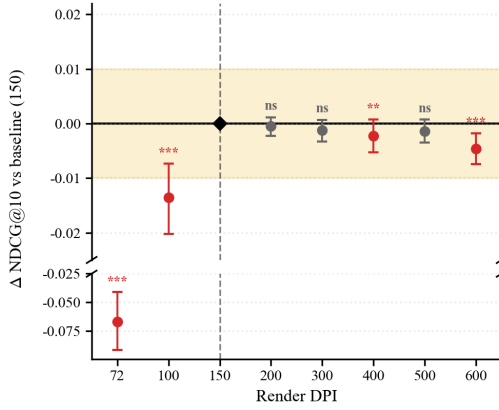


Figure 3: Embedding-DPI ablation on ViDoRe V3: mean paired $\Delta\text{NDCG@10}$ (vs. 150 DPI) against render DPI, 95% bootstrap CIs, shaded ± 0.01 two one-sided tests equivalence band (***/**/*ns significance).

fixed-protocol ablation on ViDoRe V3 (single index, identical query set).^{4 5} Across the seven DPIs compared with 150 (Fig. 3), only 72 and 100 fall outside the ± 0.01 NDCG@10 band (Cohen’s $d = -0.33$ and -0.10 ; both $p_{\text{Holm}} < 10^{-29}$), so lower DPI costs measurable quality. Every higher DPI (200–600) is equivalent to 150 (equivalence-test CI inside ± 0.01 ; effect sizes $|d| < 0.05$) but costs more per page to ingest. PULSAR therefore uses 150 DPI.

5.2 Pooling and quantisation ablation

Lower vector search latency lets the same vector-database capacity serve more concurrent searches at unchanged retrieval quality, so we target the lowest latency that holds quality. Exact MaxSim over billions of patch vectors is the dominant per-query cost. We run a ten-configuration sweep crossing pooling strategy with binary quantisation, taking the faithful no-pooling no-quantisation original configuration as the paired baseline and gating on a small effect size ($|d| < 0.1$).⁶

⁴ColQwen3-4B on ViDoRe V3 ($n=14,514$ queries). Per-query paired Wilcoxon with Holm correction; cluster-bootstrapped 95% CIs over the eight ViDoRe V3 sub-corpora. For the DPI ablation we declare practical equivalence to 150 when the TOST 90% CI on the mean paired $\Delta\text{NDCG@10}$ falls inside ± 0.01 (NDCG points); Cohen’s d (mean Δ over its paired SD) is reported as the accompanying effect size.

⁵Pages are re-rasterised from source PDFs at each target DPI. ColQwen3’s default `max_num_visual_tokens=1280` caps input at ≈ 1 MP (≈ 100 DPI on A4); we raise it to 65,536 so each render is seen at full resolution. The plateau above 150 DPI is therefore a property of ColQwen3 on these inputs.

⁶Unlike the DPI ablation (§5.1), this is not an equivalence question: pooling and quantisation trade retrieval quality for

Cfg	$\Delta\text{NDCG@10}$	$\Delta\text{Rec.}$	ms	$\times$
O (baseline)	0	0	752.7	1.0
O_q	+0.0004	+0.0001	439.4	1.7
H+O	−0.0002	−0.0001	429.8	1.8
H+ O_{qp}	+0.0001	−0.0002	257.1	2.9
M+O	−0.0014	−0.0023**	216.2	3.5
M+H	−0.0077***	−0.0065***	185.3	4.1
M+ O_{qp}	−0.0026	−0.0043***	170.9	4.4
H	−0.0059***	−0.0037***	135.4	5.6
H_q	−0.0061***	−0.0039***	73.9	10.2
M+H_{qp} *	−0.0088***	−0.0074***	49.9	15.1

Table 1: ViDoRe V3 sweep over pooling and binary quantisation, ten configurations. O/M/H are no/mean/hierarchical pooling; X+Y is X-pooled prefetch, Y-pooled rerank; q marks in-place quantisation, qp prefetch-only. Δ columns are paired differences vs. 0; ms is median vector search time; $\times$ is speedup over 0. *** $p_{\text{Holm}} < .001$, ** $< .01$; * is production.

Table 1 summarises the sweep. All nine non-baseline configurations pass the quality gate, so the choice reduces to latency. Latencies are medians over the same $n=14,514$ ViDoRe V3 queries, each issued single-stream against a single-replica vector database (Table 4) holding all ViDoRe V3 pages.⁷ M+ H_{qp} (mean-pooled binary-quantised HNSW prefetch with hierarchical-pooled rerank) gives the lowest median vector search latency at 49.9 ms (IQR 22.7 ms), a $15.1\times$ speedup over the 752.7 ms baseline (IQR 472.7 ms). The quality cost ($\Delta\text{NDCG@10} = -0.0088$, $\Delta\text{Recall@10} = -0.0074$; Cohen’s $d = -0.09$ and -0.06) clears the $|d| < 0.1$ gate, so PULSAR runs M+ H_{qp} .

The speedup comes from two effects that compose. *Pooling* shrinks the per-page vector count the reranker scores; hierarchical pooling alone (H) already gives $5.6\times$. *Binary quantisation* shrinks the per-comparison cost (O_q : $1.7\times$). Stacked, quantised hierarchical rerank (H_q) reaches $10.2\times$, and adding a quantised mean-pooled prefetch in front (M+ H_{qp}) takes it to $15.1\times$.⁸

On the chosen M+ H_{qp} index we also swept the two Qdrant search knobs. `prefetch_limit` rises in quality to a knee at 200 and is flat above; `hnsw_ef_search` has no measurable quality ef-

cost, so rather than certify parity we bound the acceptable loss by effect size on both NDCG@10 and Recall@10, and let ingest or serving cost break the tie.

⁷Pages are indexed at 150 DPI; searches use `hnsw_ef_search=128`, with `prefetch_limit=200` for the two-stage configurations.

⁸The prefetch only pays off once quantised: unquantised (M+H, $4.1\times$) it is slower than hierarchical rerank alone (H, $5.6\times$), because the prefetch HNSW lookup itself costs more than it saves the reranker.

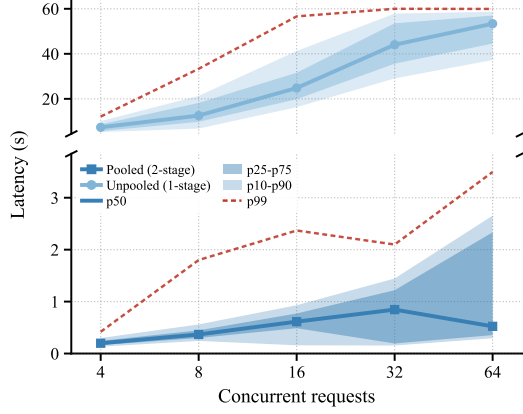


Figure 4: Per-query latency vs. concurrent requests: p50 line, p25–p75 and p10–p90 bands, p99 dashed. Pooled stays sub-second; unpooled runs an order of magnitude higher. Throughput: Fig. 5.

fect across the swept range (Friedman $p=0.14$; Table 5). PULSAR serves `prefetch_limit=200` and `hnsw_ef_search=64`.

5.3 Storage and latency costs

We validate the sweep’s pooled two-stage choice (§3.4, §5.2) on the deployment: storage and latency against an unpooled index on the dev slice (3,597 documents, 106,074 pages). The pooled $M+H_{qp}$ index keeps only the quantised mean-pooled prefetch vectors in RAM, cutting the RAM footprint $14\times$ against the unpooled 0 baseline (Table 6). At production scale the full per-page vectors would exceed cluster RAM, so an unpooled index cannot be served live at all.

On a single query the pooled index answers in a median 175 ms (95% CI 165–186) against 1.9 s for unpooled, a $10.9\times$ gap that widens to $13\times$ at p95 and $20\times$ at p99.⁹ As load rises the pooled index scales: its throughput climbs to ≈ 47 QPS at a sub-second median, every request completing. The unpooled index does not: its throughput stays flat near 0.5 QPS while latency runs to 53 s, and it drops over half its requests at 64-way (Figs. 4 and 5). In production, the same pooled index (Table 3) holds a 156 ms median vector-search latency over

⁹Load tests run over the 106,074-page dev slice on identical hardware. Both indexes retrieve by HNSW ANN. The latency/throughput curve is single-client at 1–4-way and distributed (8 pods) at 8–64-way, with completion/timeout accounting; the single-query result uses fixed sample counts (pooled $n=300$, unpooled $n=100$) after a 60 s warmup, with bootstrap 95% CIs. The load test uses the dev Qdrant (Table 4); production’s stack has $\sim 2\times$ the Qdrant CPU and disk IOPS, so these curves are a conservative lower bound on production latency.

a rolling 7-day window ($n=109,370$ calls), with a concurrency-driven tail (p95 0.6 s, p99 1.5 s) under a load that peaks at 62 QPS.

5.4 Ingestion cost

On the dev slice, we estimate ingestion cost from call counts priced at list rates, not from metered cloud spend. The OCR+verbalisation baseline runs one OCR+layout call per page plus one VLM call per figure; PULSAR runs one ColQwen3-4B forward pass per page. On this basis the baseline’s per-page ingest cost is $\approx 20\times$ PULSAR’s.¹⁰

5.5 Ingestion latency and refresh lag

Per-document end-to-end ingestion latency on the dev cluster (Table 4), measured across the dev slice, follows a rolling median $T_{\text{med}}(p) \approx 2.9 + 1.41 p$ seconds (Fig. 6): a ~ 3 s steady-state per-document overhead (object fetch, document open, index upsert) plus ≈ 1.4 s per page.

Freshness comes from skip-aware ingestion (§3.3.2): each sync re-embeds only the pages that changed, not the whole corpus, so refresh lag tracks edited-page count, not document length. A single-page edit to a multi-hundred-page deck is searchable in seconds. Over the 7-day window the pipeline cleared 50,368 changed pages, but the change rate is bursty: the busiest hour alone cleared 5,491. Change-to-searchable lag had a median of 2.0 s and a p95 of 43 s; in that busiest hour p95 held at 63 s and the worst case at 328 s. Priority scheduling (§3.3.1) runs live-deal uploads continuously and background deal material on an hourly cadence, keeping analyst uploads ahead of the background refresh.

5.6 Answer-time DPI ablation

With embedding DPI locked at 150 for retrieval, we run an answer-time DPI ablation on two firm investment-committee decks (516 questions in total; Table 8). Both are firm-authored IC packs with the chart, table, and figure density typical of production traffic. Only the page image served to the VLM at answer time varied ($\{150, 400, 500\}$ DPI). For each page, a large language model (LLM) generated three questions on its chart, table, and figure content; SMEs validated the questions and

¹⁰The baseline sent 143,934 figures for verbalisation across the slice; only 38,868 (27%) yielded a description the downstream filter kept. Every call is still paid for, so nearly three-quarters of the per-figure VLM cost buys output that is discarded.

authored the gold answers. The same question set was reused at every DPI. SMEs labelled each system answer as correct or as a page-legibility error: an unreadable page image at that DPI, not generation drift on a faithful page (Ji et al., 2023).

Page-legibility errors fall monotonically with DPI on both decks, reaching zero at 500, so PULSAR serves 500 DPI at answer time.¹¹

5.7 PULSAR vs the baseline

We compare PULSAR end to end against the OCR+verbalisation baseline on answer quality. PULSAR scores higher on Completeness and both context- and answer-fact recall at every $k \in \{1, 3, 5, 10\}$ (Table 2). The evaluation set is $n=75$ SME-authored investment questions with SME-authored gold answers (categorised in App. H); both retrieval paths run the same questions and generate answers with the same answer-time VLM, so only the indexing path differs (verbalised text chunks vs. page images). An LLM judge (validated against blinded SME re-scoring; App. I) scores each system’s output against the gold. The two paths index different units, so we report answer-level, modality-independent metrics, not retrieval rank.

To separate run-to-run model noise from sampling uncertainty, we cluster by deal (the correlated unit) and report in the bottom block of Table 2 the deal-weighted PULSAR–baseline delta with a

95% bootstrap CI over the 7 deals.¹² Every interval excludes 0 at every k ; PULSAR beats the baseline on all seven deals in all twelve metric-by- k cells (exact Wilcoxon signed-rank $p=.016$, the floor at $n=7$), and the advantage survives leave-one-deal-out.

The advantage is retrieval-led. ContextFactRecall is $1.9\text{--}2.3\times$ the baseline from $k=3$ up, and larger still at $k=1$, where the baseline retrieves almost nothing (Table 2). This is consistent with vision-first retrieval preserving the chart values and legend assignments condensed during verbalisation. AnswerFactRecall inherits this gap: it is widest at low k , where PULSAR reaches roughly $5\times$ the baseline (.49 vs .10 at $k=1$), and narrows as the baseline catches up. At the production $k=5$ (§3.2), PULSAR already exceeds the baseline’s $k=10$ on every metric.

6 Conclusion

PULSAR runs vision-first RAG in production at firm scale. Two choices keep it affordable: decoupling embedding DPI (150) from answer-time DPI (500), and a pooled two-stage late-interaction index that cuts median vector-search latency $15.1\times$ and sustains $\approx 88\times$ an unpooled index’s throughput under load, at under 0.01 NDCG@10 and Recall@10 loss. Ingestion is $\approx 20\times$ cheaper per page than the OCR+verbalisation baseline, and answer-fact recall more than doubles.

¹²The $\pm$ in Table 2 is temperature-0 run-to-run non-determinism in the answer-time VLM and LLM judge, not sampling error.

Metric	$k=1$	$k=3$	$k=5$	$k=10$
<i>OCR+verbalisation baseline</i>				
Completeness (1–5)	$1.68 \pm .02$	$2.34 \pm .04$	$2.55 \pm .03$	$2.85 \pm .04$
ContextFactRecall	$.134 \pm .005$	$.322 \pm .004$	$.388 \pm .013$	$.507 \pm .008$
AnswerFactRecall	$.102 \pm .002$	$.226 \pm .010$	$.271 \pm .005$	$.355 \pm .003$
<i>PULSAR (vision-first)</i>				
Completeness (1–5)	$3.41 \pm .06$	$3.89 \pm .07$	$4.15 \pm .06$	$4.14 \pm .04$
ContextFactRecall	$.587 \pm .004$	$.745 \pm .006$	$.874 \pm .006$	$.949 \pm .002$
AnswerFactRecall	$.492 \pm .007$	$.650 \pm .021$	$.733 \pm .009$	$.729 \pm .015$
Δ deal-weighted (PULSAR – baseline), 95% CI over 7 deals				
Completeness (1–5)	+1.90 [1.18, 2.62]	+1.75 [1.26, 2.24]	+1.77 [1.28, 2.25]	+1.51 [0.78, 2.25]
ContextFactRecall	+0.51 [0.35, 0.66]	+0.47 [0.28, 0.66]	+0.55 [0.37, 0.73]	+0.50 [0.29, 0.72]
AnswerFactRecall	+0.43 [0.27, 0.59]	+0.47 [0.33, 0.62]	+0.52 [0.37, 0.66]	+0.44 [0.25, 0.64]

Table 2: PULSAR vs the OCR+verbalisation baseline on $n=75$ SME-authored questions (§5.7); cells are mean $\pm$ std over 5 passes. Completeness is a 1–5 LLM judgement of the system’s answer against the gold answer; Context/AnswerFactRecall are the shares of gold facts in the retrieved context and in the answer. Bold marks PULSAR higher at the same k .

Limitations

Late interaction stays heavier than single-vector retrieval. Even with hierarchical pooling and the two-stage index (§3.4), each page is still stored and scored as many pooled vectors rather than one, and MaxSim remains costlier than a single dot product. These optimisations cut late interaction’s footprint and scoring cost but do not erase this overhead: it stays heavier than single-vector retrieval.

Findings are tied to this corpus, baseline, and backbone. All production measurements come from one firm’s chart- and table-heavy private-markets deal documents, run on ColQwen3-4B. The DPI sweet spot, ingestion-cost gap, and factuality margins may shift on corpora with different chart density or page complexity, on other vision-first backbones, and against text-extraction stacks that preserve more chart detail than the OCR+verbalisation baseline here. Our retrieval-quality evidence for the index design, the embedding-DPI and pooling/quantisation equivalence tests (§5.1, §5.2), is measured on the public ViDoRe V3 benchmark rather than the firm corpus; we have not confirmed these equivalences hold on the firm’s own documents.

Ingest-cost estimate is ingestion-only. The $\approx 20\times$ gap (§5.4) covers ingestion calls only; standing serving and answer-time inference are excluded.

Answer-time DPI ablation uses SME ground truth and LLM-generated questions. The 516-question DPI result (§5.6) checks each DPI’s answers against factual ground truth the finance SMEs authored ahead of time. Labels are value-level matches (e.g. does the answer’s stated internal rate of return equal the SME’s), not subjective quality judgements. The question-generating LLM is in the same model family as the answer-time VLM, so both may share the same visual blind spots. The protocol isolates the answer-generation step: each question is generated from a specific page, which is fed directly to the VLM, so retrieval is not exercised. It does not speak to end-to-end behaviour, where retrieval can also fail. The 516 questions cluster on two decks, so the effective sample size for between-deck generalisation is two; the result speaks to behaviour on IC-pack-style pages, not arbitrary documents.

We compare against one baseline, not alternative vision-first stacks. The OCR+verbalisation baseline vs vision-first comparison in §5.7 runs

75 SME-authored questions through both systems and scores three LLM-judged metrics against SME-authored gold answers, repeated over 5 independent passes. A blind 20% subsample re-scored by three domain experts tracks the judge (weighted $\kappa \geq 0.79$) and independently favours PULSAR (App. I), so the comparison is human-validated, not judge-only. We do not compare against alternative vision-first stacks (e.g. OCR-free document transformers (Kim et al., 2022; Hu et al., 2024)) or generate-and-encode hybrids (Nguyen et al., 2025), so we do not claim late-interaction page-image retrieval is the best approach for this task.

Ethical Considerations

Decision impact. PULSAR informs investment decisions, so even a low residual error rate carries downstream consequences. It is surfaced to analysts as an advisory aid, and outputs that propagate into client-facing artefacts go through human review. We do not recommend deploying vision-first RAG as an unsupervised decision-maker on investment material.

Data. Retrieval quality is benchmarked on the public ViDoRe V3 suite. The production deployment, the end-to-end answer evaluation, and the load and ingestion measurements run on Mubadala’s confidential deal documents, which are not released and are not used to train or fine-tune any model: PULSAR runs a frozen pretrained backbone, and all indexing and evaluation of that corpus happen within the firm’s own environment under its data-governance controls. The evaluation questions and gold answers were authored by SMEs already authorised to access the underlying documents.

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A Configuration and Hardware Setup

Table 3 lists the index and serving configuration and Table 4 the Kubernetes cluster sizing, each across the benchmark, dev, and prod settings. The benchmark columns give the settings behind the §5.1 and §5.2 ablations; the dev and prod columns describe the deployed system.

B Priority-aware ingestion scheduling

Scheduling (§3.3.1) splits across two layers.

Document-level (dispatcher). The dispatcher pulls from the highest non-empty lane and a higher-priority arrival evicts any in-flight lower-priority job. Each page is stamped with its page hash at upsert (not the document hash, which is stamped only once the whole document is done, §3.3.2), so on redelivery the evicted job resumes from the last durable page; wasted work is bounded by the in-flight batch, not document length.

Request-level (vLLM). Within a lane, each embedding request is stamped with a numeric priority

¹³The pixel (max_pixels) and sequence-length (max_model_len) caps are set so a standard A4 page embeds without clipping.

Parameter	Benchmark	Dev	Prod
Backbone	ColQwen3-4B (Qwen3-VL backbone)		
Model dtype	bfloat16		
Vector dim D	320		
patch_size / merge_size	16 / 2		
Distance metric	cosine (MaxSim multivector)		
max_num_visual_tokens	65,536	4,096	
Image cap max_pixels	67,108,864	4,194,304	
Embedding/index DPI	swept	150	
Stored/served DPI	n/a	500	
Vector DB engine	Qdrant 1.16	Qdrant 1.17	
HNSW graph degree m	16		
HNSW ef_construct	100		
Index configuration	swept	M+H _{qp}	
Prefetch pooling	swept	mean row + column	
Rerank pooling	swept	hierarchical, pool factor 3	
Quantisation	swept	binary, prefetch-only	
hnsw_ef_search	swept	64	
prefetch_limit	swept	200	
exact	false		
indexed_only	false	true	
Serving engine	transformers	vLLM 0.19.1	
max_model_len	n/a	3,072	
GPU KV cache size	n/a	4,560 tokens	
cpu_offload_gb	n/a	0.3	
KV cache dtype	n/a	bf16	
gpu_memory_utilization	n/a	0.89	
enforce_eager	n/a	true	
max_num_batched_tokens	n/a	8,192	
max_num_seqs	n/a	16	
Returned pages	@ 10 (eval)	top- $k=5$	
Query issue	single-stream	concurrent	

Table 3: Index/serving configuration for the offline benchmark and the deployed system (dev and prod, both full-res A4 via CPU weight offload).¹³ *swept* marks parameters varied in the §5.1 and §5.2 ablations; cluster sizing is in Table 4. Binary quantisation rescores at 2× oversampling.

Component	Benchmark	Dev	Prod
Transformers inference	1× (1/4, 8/16)	–	–
vLLM inference	–	2× (2/4, 12/20)	4× (2/4, 16/24)
GPU node	NV18ads_A10_v5 (all)		
colpali-inference	–	2× (1/4, 6/12)	4× (1/4, 6/12)
Doc indexer	–	2× (1/2, 6/24)	4× (1/2, 6/24)
Vector DB	1× (6/7.5, 25/28)	3× (6/10, 50/100)	4× (10/14, 64/100)
Vector DB PVC	512 GiB (2,300)	512 GiB P20 (2,300)	1 TiB P30 (5,000)

Table 4: Kubernetes sizing (replicas × CPU/mem, requests/limits), persistent-volume managed-disk tiers with base provisioned IOPS in parentheses; each inference pod gets one NV18ads_A10_v5 (a fractional-A10 SKU), so the GPU count equals the inference replica count. The benchmark ran as Azure ML pipeline jobs (AKS-backed) on a single-replica stack (512 GiB managed-premium Qdrant PVC): GPU work on NV18ads_A10_v5, CPU work on Standard_D8s_v3. It uses no separate doc-indexer component (–).

derived from remaining pages, so the GPU drains shortest remaining processing time (SRPT) first and short documents do not queue behind long ones. A time-in-flight term ages low-priority work up so it cannot starve. Search queries enter at top priority, so foreground search is never blocked by background embedding.

C Search-parameter Sweeps

Table 5 gives the full `prefetch_limit` and `hnsw_ef_search` sweeps behind the search-knob choices in §5.2.

Setting	$\Delta\text{NDCG@10}$	p_{50}	p_{90}	p_{95}	p_{99}
<i>prefetch_limit (baseline 200)</i>					
25	−.032***	36	57	66	85
50	−.016***	38	59	68	88
100	−.006***	43	66	76	105
150	−.002***	46	71	81	105
200*	0	50	77	88	110
250	ns	54	81	93	121
300	ns	57	86	97	123
400	ns	62	92	104	131
800	ns	87	133	153	193
<i>hnsw_ef_search (baseline 128)</i>					
64*	ns	50	79	90	113
128	0	53	82	94	119
256	ns	53	82	94	120
512	ns	54	81	91	113

Table 5: Search-parameter sweeps on ViDoRe V3 (§5.2), on the selected M+H_{qp} index at 150 DPI, $n=14,514$ queries. $\Delta\text{NDCG@10}$ is the mean paired difference vs. each sweep’s baseline (Wilcoxon + Holm within sweep; *ns* = indistinguishable on NDCG@10). p_{50} – p_{99} are per-query Qdrant server-side latencies (ms). *** $p_{\text{Holm}} < .001$. * marks the deployed setting.

D Index Footprint

Table 6 gives the storage footprint summarised in §5.3, on the 106,074-page dev slice.

	Pooled (M+H_{qp})	Unpooled (0)	× smaller
RAM-resident vectors	12 GB	169 GB	14
On disk (total)	69 GB	166 GB	2.4

Table 6: Index footprint of the pooled two-stage M+H_{qp} configuration against the unoptimised 0 baseline (Table 1) on the 106,074-page dev slice. Pooled keeps only the quantised mean-pooled prefetch vectors resident and pushes the hierarchical rerank vectors to disk; unpooled keeps every full per-page multivector resident. GB figures are approximate.

E Retrieval Throughput Under Load

Fig. 5 reports the throughput behind the latency comparison in §5.3: throughput over a 4–64-way sweep for both indexes at corpus scale.

We also isolate per-query latency from queueing with a single-concurrency run (Table 7): warm, no concurrent load, fixed sample counts, bootstrap

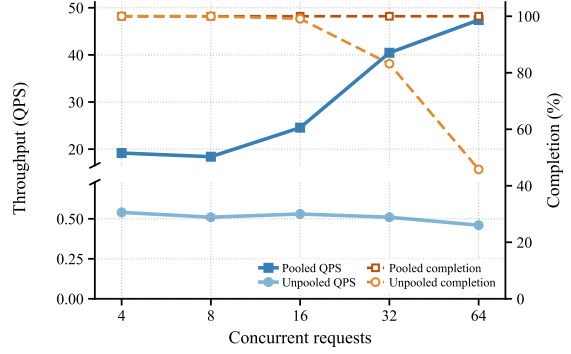


Figure 5: Throughput (left, QPS) and request completion (right, %) against concurrent requests, 106,074-page dev slice. Pooled climbs to ≈ 47 QPS and holds 100% completion; the unpooled index tops out at 0.54 QPS ($\approx 88\times$ lower) and its completion collapses under load, from 100% to 46% at 64-way, because each full-multivector scan is so costly the system saturates at one-way concurrency. Per-query latency in Fig. 4.

95% CIs. The pooled index answers in a median 175 ms against 1.9 s for unpooled, and the gap widens through the tail.

	p50	p90	p95	p99
Pooled	175	387	546	686
Unpooled	1,907	4,628	6,966	13,639

Table 7: Single-query latency (ms) at concurrency 1 over the 106,074-page dev slice, warmup discarded (pooled $n=300$, unpooled $n=100$). Bootstrap 95% CIs: pooled p50 165–186, p95 405–621; unpooled p50 1,860–1,966, p95 4,351–8,748.

F Ingestion Latency

Fig. 6 plots the per-document ingestion latency summarised in §5.5, on the dev slice.

G Answer-time DPI Legibility Errors

Table 8 gives the per-deck page-legibility error counts behind the answer-time DPI ablation in §5.6.

Deck	Pages	Qs	Vis. 150	Vis. 400	Vis. 500
Deck A	94	282	18	8	0
Deck B	78	234	21	7	0

Table 8: Page-legibility errors on two firm investment-committee decks as answer-time DPI varies (§5.6).

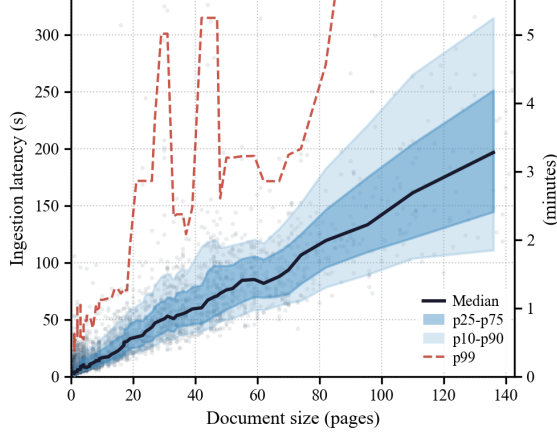


Figure 6: End-to-end ingestion latency vs. document length (dev cluster, 2 indexer pods $\times$ concurrency 2): rolling median, IQR, p10–p90, and p99; x-axis clipped at the 97th percentile (143 pages).

H Evaluation Question Taxonomy

We categorise the $n=75$ head-to-head evaluation questions (§5.7) on two axes, assigning each question one label per axis. Examples are real questions with company, deal, and competitor names removed.

Financial subject. (i) Operating and financial metrics (22; 29.3%): “What is the projected trajectory for EBITDA and margins over the forecast period for the company?” (ii) Market and competitive context (20; 26.7%): “What does the slide indicate about the projected growth and CAGR of the overall sector market over the forecast period?” (iii) Valuation and returns (14; 18.7%): “What overall company valuation is implied by the proposed entry share price?” (iv) Capital structure and funding (12; 16.0%): “What is the aggregate amount and denomination of the rated senior and junior debt facilities?” (v) Segment and revenue mix (7; 9.3%): “How is the mix of contracted and non-contracted revenue projected to change over the forecast period?”

Question operation. (i) Value extraction (54; 72.0%): “What is the reported daily active user count for the platform?” (ii) Cross-entity or benchmark comparison (13; 17.3%): “What are a peer’s total payment volume and EBITDA margin, and how does its profitability measure up against the company’s?” (iii) Temporal classification, reported vs projected (8; 10.7%): “Which year does the company report actual net revenue and EBITDA margin figures, rather than a forward projection?”

I Human Validation of the LLM Judge

To check that the LLM-judged head-to-head (§5.7) reflects human judgement, three domain-expert SMEs blindly re-scored a deal-stratified 20% subsample (15 questions, 64 atomic gold facts) of the pass-1 answers. The two systems were shown as “Answer A”/“Answer B” in randomised order with modality cues removed, so raters scored content only, on the same two metrics (Completeness 1–5, AnswerFactRecall 0/1 per fact).

Agreement with the judge. Per rater, Completeness tracks the judge at Pearson $r=0.83$ – 0.86 and quadratic-weighted $\kappa=0.79$ – 0.81 , and AnswerFactRecall at $r=0.94$ – 0.98 (Table 9). Agreement between raters is “almost perfect” (Landis–Koch): pairwise weighted $\kappa=0.84$ – 0.96 on Completeness, and Fleiss $\kappa=0.89$ on AnswerFactRecall over 128 fact-answer judgements (64 facts $\times$ 2 systems).

Direction confirmed. All three SMEs independently rank PULSAR above the baseline, each by a margin at least as large as the judge’s (Completeness +1.07 to +1.60 vs the judge’s +1.07; AnswerFactRecall +0.32 to +0.41 vs +0.32), so the judge is conservative about the gap rather than inflating it; where human and judge diverge, the human penalises PULSAR’s own hallucinations harder.

Rater	Completeness		FactRec.
	r	wt. κ	r
SME1	0.86	0.81	0.98
SME2	0.84	0.80	0.98
SME3	0.83	0.79	0.94
median rater	0.85	0.81	–

Table 9: Agreement of three blind SME raters with the LLM judge on the 20% subsample (§5.7): Pearson r and quadratic-weighted κ for Completeness, r for AnswerFactRecall. The *median rater* correlates the per-answer median of the three SMEs, taken as one consensus rating, against the judge.